Actors are the primary building blocks in the actor model. All the components in the example application are actors, shown in figure 1.16. An actor is a lightweight process with only four core operations: create, send, designate the next behavior, and supervise. All of these operations are asynchronous.

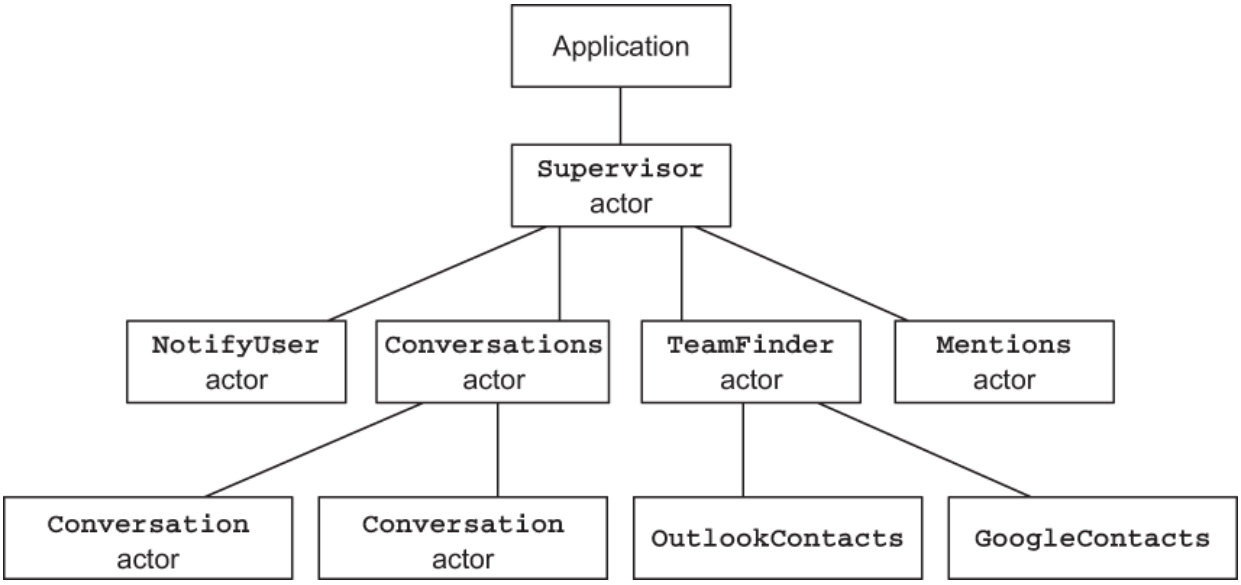


Figure 1.16 Application components

The actor model—not new

The actor model has been around for quite a while; the idea was introduced in 1973 by Carl Hewitt, Peter Bishop, and Richard Steiger. The Erlang language and its Open Telecom Platform (OTP) middleware libraries, developed by Ericsson around 1986, support the actor model and have been used to build massively scalable systems with requirements for high availability. An example of the success of Erlang is the AXD 301 switch product, which achieves a reliability of 99.9999999%, also known as *nine nines* reliability. The actor model implementation in Akka differs in a couple of details from the Erlang implementation but has been heavily influenced by Erlang and shares many of its concepts.